

Effect of alpha-tubulin acetylation on the doublet microtubule structure

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Abstract

Acetylation of α -tubulin at the lysine 40 residue (α K40) by ATAT1/MEC-17 acetyltransferase modulates microtubule properties and occurs in most eukaryotic cells. Previous literatures suggest that acetylated microtubules are more stable and damage-resistant. α K40 acetylation is the only known microtubule luminal post-translational modification site. The luminal location suggests that the modification tunes the lateral interaction of protofilaments inside the microtubule. In this study, we examined the effect of tubulin acetylation on the doublet microtubule in the cilia of *Tetrahymena thermophila* using a combination of cryo-electron microscopy, molecular dynamics, and mass spectrometry. We found that α K40 acetylation exerts a small-scale effect on the doublet microtubule structure and stability by influencing the lateral rotational angle. In addition, comparative mass spectrometry revealed a link between α K40 acetylation and phosphorylation in cilia.

eLife assessment

This **fundamental** study employs a combination of cryo-electron microscopy, molecular dynamics, and mass spectrometry to elucidate the role of α -tubulin acetylation at the luminal lysine 40 residue (α K40) within the cilium. **Compelling** evidence shows α K40 acetylation to impact the structure and stability of doublet microtubules in cilia by affecting the lateral rotational angle. The work will be of relevance to those interested in cytoskeleton and structural biology.

Introduction

Cilia have diverse roles in cell motility, sensory functions, signalling, and growth control. Motile cilia drive the flow of fluid, including human sperm (Lehti & Sironen, 2017 ) , mucus clearance in the respiratory tract (Bustamante-Marin & Ostrowski, 2017 ) , and cerebrospinal fluid circulation (Djenoune & Wyart, 2017 ) . Non-motile primary cilia in rod cells are critical for transmitting chemical signals converted from light (Sjostrand, 1953 ) . In kidney epithelial cells, cilia function

as mechanosensors for transmitting fluid flow signals into signalling pathways. At the core, motile and primary cilia share the same cytoskeletal framework, the axoneme (Porter & Sale, 2000 [↗](#)), composed of a bundle of nine outer doublet microtubules (DMTs). Each DMT is composed of protofilaments (PFs) of tubulins that form into a hollow cylinder A-tubule and an incomplete cylinder B-tubule (Fig. 1A [↗](#)). Inside the DMT lumen, a weaving network of microtubule inner proteins (MIPs) stabilizes the DMT structure (Ichikawa et al., 2019 [↗](#); Ichikawa et al., 2017 [↗](#); Ma et al., 2019 [↗](#)).

Tubulins undergo several highly conserved post-translational modifications (PTMs) that collectively represent the so-called ‘tubulin code’, in which PTMs modulate microtubule properties directly or indirectly through the binding of microtubule-associated proteins (MAPs). The common and well-studied tubulin PTMs are phosphorylation, deetyrosination, glutamylation, glycylation and acetylation (Wloga et al., 2017 [↗](#)). DMT contains a unique signature of PTMs including cilia specific PTMs such as glycylation. For example, the B-tubule of DMT is enriched with glutamylated (Lechtreck & Geimer, 2000 [↗](#)) and deetyrosinated (Johnson, 1998 [↗](#)) tubulins, while A-tubule tubulins are mostly unmodified. PTM is a fine tune for ciliary function rather than a biphasic switch. Glutamylation could regulate inner dynein arm activities, which control the ciliary waveform (Kubo et al., 2010 [↗](#); Lechtreck & Geimer, 2000 [↗](#); Suryavanshi et al., 2010 [↗](#)). Hyperglutamylation due to depletion of deglutamylases can improve intraflagellar transport in ift88-deficient zebrafish (Pathak et al., 2014 [↗](#)). Lack of glycylation causes abnormal pre-powerstroke and post-powerstroke conformations of dynein arms in mouse sperm (Gadadhar et al., 2021 [↗](#)). The loss of glycylation sometimes leads to an increase in glutamylation, possibly because the two PTMs compete for the same set of modification sites (glutamic acids) on tubulins (Kubo et al., 2015 [↗](#); Rogowski et al., 2019 [↗](#); Rogowski et al., 2010a [↗](#); Dorota Wloga et al., 2009 [↗](#)). These observations suggest that there is an interplay between PTMs and cilia properties.

One of the most intriguing PTMs in cilia is acetylation, which occurs inside the lumen of DMT on the lysine 40 residue of α -tubulins (α K40). α K40 acetylation in *Chlamydomonas reinhardtii* cilia was the first identified tubulin acetylation (M & Piperno, 1987; Sw & Rosenbaum, 1983 [↗](#)). Tubulin acetylation was later found on different microtubules in cells, such as in neurons (Fukushige et al., 1999 [↗](#)). Acetylation could also take place on lysine 60 and lysine 370 residues of α -tubulin and lysine 58 of β -tubulin (Liu et al., 2015 [↗](#)). Acetylation is interesting because the α K40 loop is the only luminal PTM site and is close to the tubulin lateral interaction interface (Kaul et al., 2014 [↗](#)) (Fig. 1B [↗](#)). The α K40 loop is flexible and close to the tubulin lateral interaction interface (Eshun-Wilson et al., 2019 [↗](#); Howes et al., 2014 [↗](#)) and is almost 100% completely acetylated in cilia (Akella et al., 2010 [↗](#)). Acetylated α K40 has been shown to enhance microtubule stability and longevity *in vitro* (Schaedel et al., 2015 [↗](#)), while deacetylated microtubules decrease in rigidity and are prone to complete breakage events (Xu et al., 2017 [↗](#)). Biochemical works on reconstituted microtubules from acetylated and deacetylated tubulins suggest that α K40 acetylation directly weakens inter-PF interactions and, hence enhance flexibility and reduce mechanical fatigue of microtubule (Portran et al., 2017 [↗](#)).

The main enzyme responsible for α K40 acetylation is alpha-acetyltransferase-1 (α TAT1), also known as MEC-17 (Akella et al., 2010 [↗](#)). Deacetylation is carried out by histone deacetylase 6 (HDAC6) (Hubbert et al., 2002 [↗](#)) and nicotinamide adenine dinucleotide-dependent deacetylase sirtuin 2 (SIRT2) (North et al., 2003 [↗](#)). Mice that lack α TAT1 have defective sperm flagellar beating (Kalebic et al., 2013), and *C. elegans* become touch insensitive without α TAT1 (Akella et al., 2010 [↗](#); Shida et al., 2010 [↗](#)). Motor proteins travel preferentially on acetyl-K40 microtubules because of their higher binding affinity (Garnham & Roll-Mecak, 2012 [↗](#); Reed et al., 2006 [↗](#)). Overexpression of HDAC6 or SIRT2 could lead to short cilia (Ran et al., 2015 [↗](#); Zhou et al., 2014 [↗](#)), but the cilia-shortening effect of HDAC6 could be countered by an acetyl-K mimic α -tubulin, suggesting that it may destabilize the microtubule lattice (Cueva, Hsin, et al., 2012). The α K40 loop is functionally important, but it is flexible and disordered even in both acetylated and deacetylated reconstituted microtubules (Eshun-Wilson et al., 2019 [↗](#); Howes et al., 2014 [↗](#)). Cryo-EM and molecular dynamic

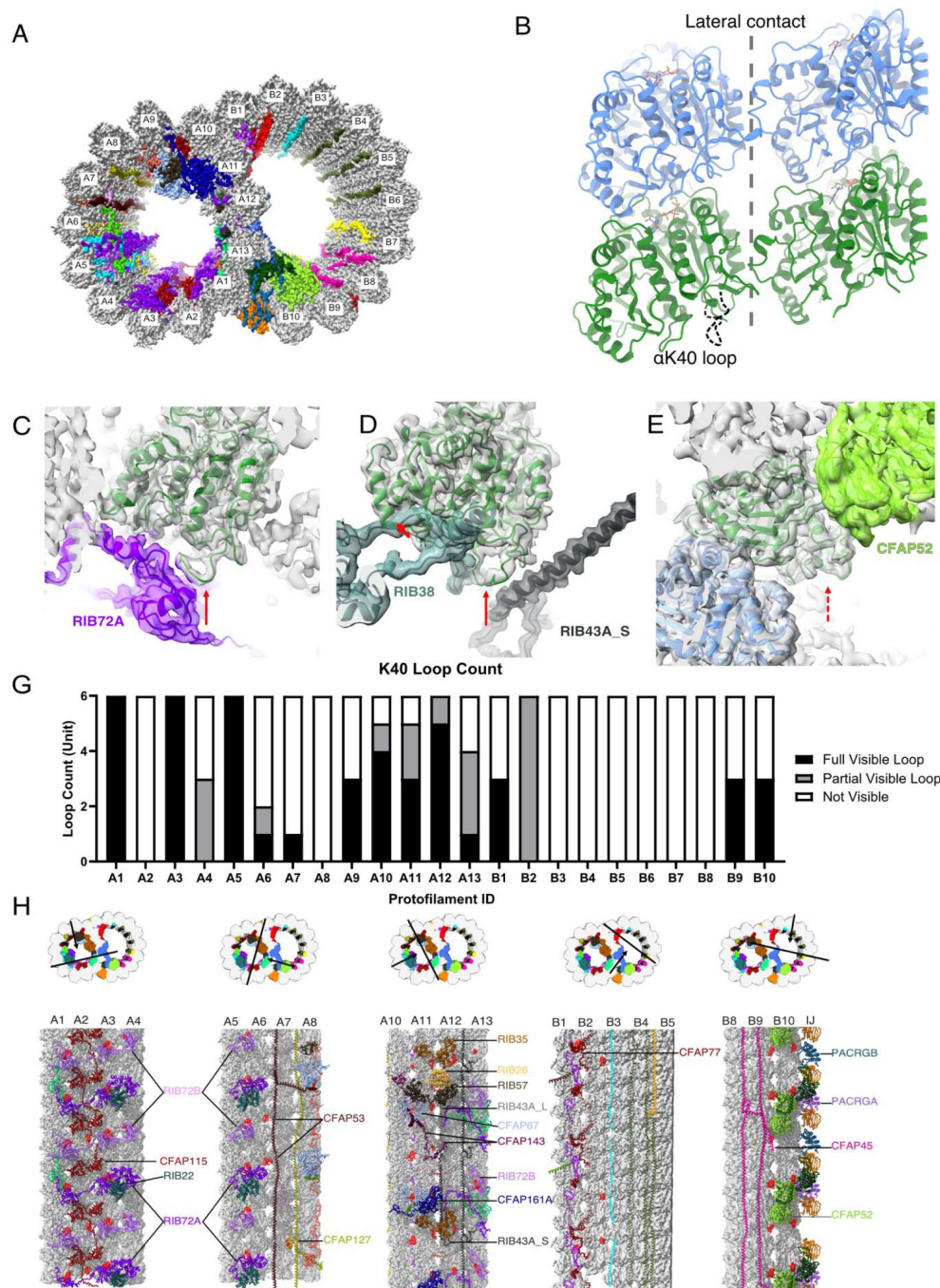


Figure 1.

The presence of the structure α K40 loops in *Tetrahymena thermophila* DMT.

(A) Surface rendering of the DMT viewed from the tip of the cilia, with MIPs colored in the 48-nm repeat cryo-EM electron density map of *Tetrahymena*. (B) Relative location of the α K40 loop (dashed black line) and the lateral contacts of tubulins. Color: α -tubulin, green; β -tubulin, blue. (C-F) Cryo-EM map and models of the fully structured α K40 loops in PF A3 (C-D) and the fully structured (E) and partially structured (F) α K40 loops in PF B10. The red arrows point to the location of the α K40 loops. (G) Bar graph showing the composition of visible full (missing no more than two residues from residues 37 to 48) and partial loops (missing 3-5 residues) in both the A- and B-tubules. (H) 48-nm repeat surface rendering of selected PFs with MIPs that interact with visible full and partial α K40 loops colored in red, indicating that α K40 loops are structured in regions with many MIPs.

studies of acetylated microtubules suggest that acetylation restricts the α K40 loop motion by disturbing the electrostatic interaction (Eshun-Wilson et al., 2019). This produces more ordered loops and stabilizes the microtubule lattice. In certain organisms, such as *C. elegans*, acetylation of α K40 would disrupt an intramonomer salt bridge from α E55 to α K40 (Cueva, Hsin, et al., 2012). Replacing α K40 with arginine or in the absence of acetyl- α K40 could result in the formation of a salt bridge from α E55 with α K40, α R40, or α H283, to change the inter-PF angle leading to elliptical microtubules and variation in the number of PFs (Cueva, Hsin, et al., 2012).

Interestingly, many α K40 loops are fully structured in the cryo-EM map of the DMT from ciliate and green algae (Khalifa et al., 2020; Ma et al., 2019). This result suggests that the α K40 loop may have an important role in recognizing and binding with different MIPs to stabilize cilia, and disruption of acetylation may disrupt the interaction between the α K40 loop and MIPs.

In this work, we aimed to clarify the role of α K40 acetylation in the assembly and stability of ciliary DMT. To answer our questions, we studied the structural effect of acetylated and non-acetylated tubulins on DMT and MIPs from *wild-type* (WT) *Tetrahymena thermophila*, mutants lacking tubulin acetyltransferase (MEC-17) and α K40-specific de-acetylation (*K40R*) (Akella et al., 2010) by a combination of cryo-EM, molecular dynamics and mass spectrometry.

Results

Acetylated α K40 loops are structured when interacting with certain MIPs

We first examined a 48-nm repeat cryo-EM density map reconstructed from native DMTs of *Tetrahymena* WT (CU428 strain) (Fig. 1A, Supplementary Fig. 1, Table 1 and 2) (Kubo et al., 2023). We identified and modeled all visible α K40 loops of α -tubulins in the map, which are known to be almost 100% acetylated (Akella et al., 2010; Gaertig et al., 1995). Most *Tetrahymena* MIPs have been localized and identified in this map (Kubo et al., 2023). In contrast to the acetylated singlet microtubule structure (Eshun-Wilson et al., 2019), we observed many fully structured α K40 full (missing no more than two residues from residues 37 to 48) and partial structure loops (missing 3-5 residues) in DMT (Fig. 1C-E). In certain PFs, including A1, A3, A5 and A12, all α K40 loops are structured. Interestingly, the structured α K40 loops are much less abundant in the B-tubule (Fig. 1F). There is a pattern in the distribution of α K40 loops: likely due to their flexibility, ordered α K40 loops are visible at positions where MIPs and tubulins interact but are difficult to resolve in places with little or no MIPs. (Fig. 1G, Supplementary Fig. 2, Supplementary Table 1). Notably, in A1, where RIB72A and RIB72B are in contact with α K40 (Fig. 1C, Supplementary Fig. 2), all α K40 loops are fully structured, in agreement with the 8-nm alternating pattern of Rib72A and Rib72B (Fig. 1G). This pattern is consistent with previous observations of the α K40 loops in the green algae *Chlamydomonas reinhardtii* DMT (Khalifa et al., 2020; Ma et al., 2019). Therefore, the B-tubule has fewer ordered α K40 loops compared to the A-tubule, probably due to fewer MIPs in the B-tubule.

A superimposition of the conformations of all fully structured α K40 loops in the DMT revealed that they adopt multiple conformations (Fig. 2A). The exact structures of α K40 loops might adapt to both the inter-PF angle and interactions with the contacting MIPs. We did not observe any conformations where acetylated α K40 was directly involved in tubulin-tubulin lateral interactions (Fig. 2A, Fig. 1C-E).

We noticed that the α K40 loop is always structured when interacting with DM10 domains of the MIPs, suggesting that DM10 is a α K40 loop interacting domain. Superimposing all seven DM10 domains (three from RIB72A, three from RIB72B and one from CFAP67) (Fig. 2B) showed that while there are some variations in other parts of DM10 domain, the region interacting with the

	WT (CU428)		K40R		MEC17-KO
Dataset	Dataset 1	Dataset 2	Dataset 1	Dataset 2	Dataset 1
Microscope	Titan Krios	Titan Krios	Titan Krios	Titan Krios	Titan Krios
Electron Detector	Gatan K3	Gatan K3	Gatan K3	Gatan K3	Gatan K3
Zero-loss filter (eV)	30	30	30	30	30
Magnification	64,000	64,000	64,000	64,000	64,000
Voltage	300	300	300	300	300
Electron exposure	45 e/A ²	45 e/A ²	73 e/A ²	45 e/A ²	45 e/A ²
Defocus range	1.0-3.0 μ m	1.0-3.0 μ m	1.0-3.0 μ m	1.0-3.0 μ m	1.0-3.0 μ m
Pixel size	1.37	1.37	1.37	1.37	1.37
Symmetry imposed	C1	C1	C1	C1	C1
Movies acquired	4,080	14,304	4,400	21,210	4283
Particles number	40,945	125,306	30,352	152,303	39,417
Global resolution (Å)	4.51	4.26	4.0	3.6	4.5
Local resolution (Å)	3.6 – 4.0		3.3 – 3.5		4.0 – 4.5

Table 1

Cryo-EM data collection refinement

	WT (CU428)	K40R	MEC17-KO
Model-to-Map fit, CCmask	0.8009	0.7961	0.6070
All-atom clashscore	16.61	13.21	57.06
Ramachandran plot			
Outliers [%]	0.14	0.14	0.14
Allowed [%]	3.70	3.60	6.13
Favored [%]	96.16	96.26	93.73
Rotamer outliers [%]	0.03	0.04	0.02
Cbeta deviations [%]	0.00	0.00	0.01
Cis-proline [%]	4.56	4.56	4.54
Cis-general [%]	0.01	0.01	0.03
Twisted proline [%]	0.17	0.18	0.13
Twisted general [%]	0.02	0.01	0.02

Table 2

Refinement statistics of WT, K40R and MEC17-KO 48nm models.

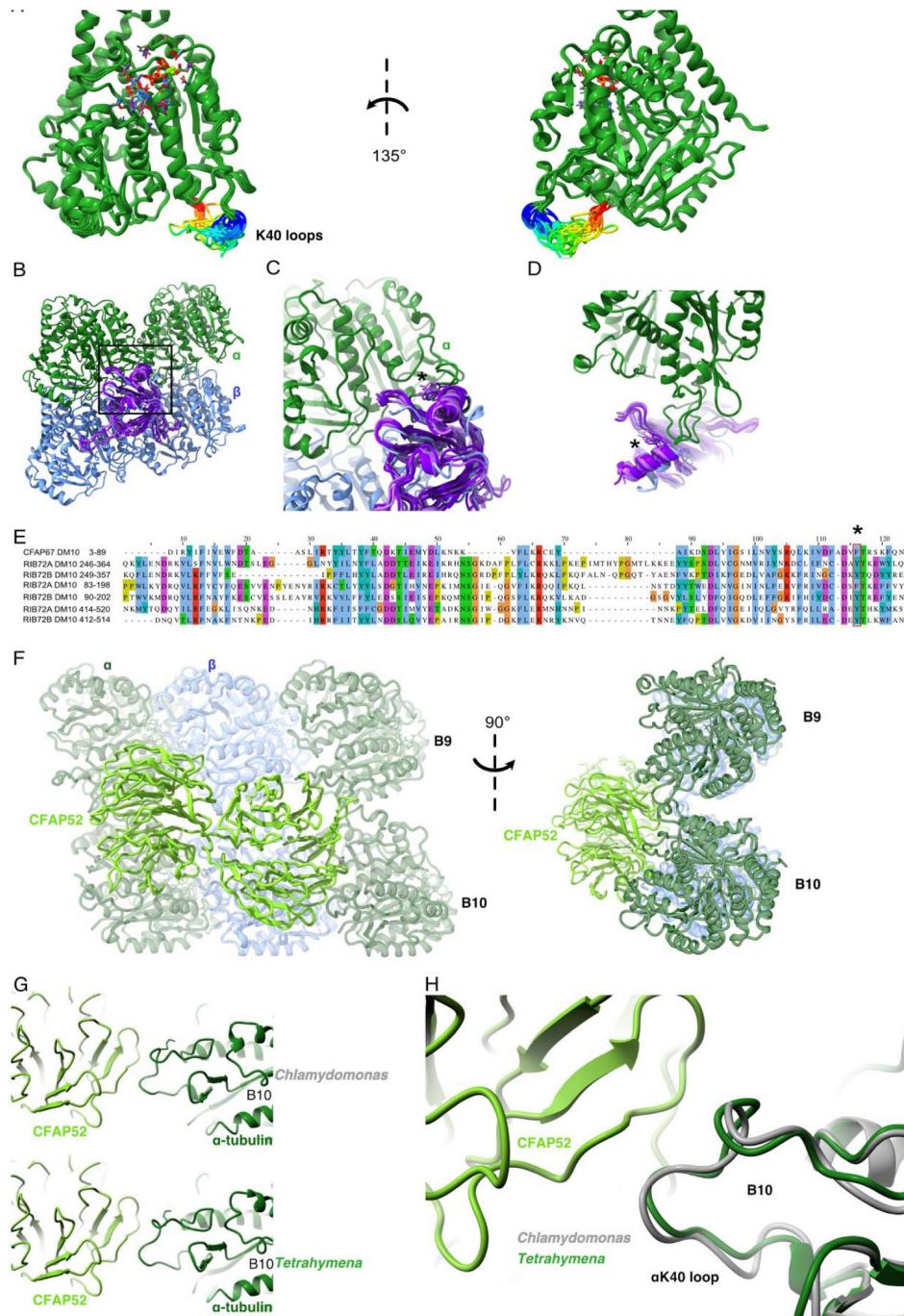


Figure 2

Comparison of αK40 loop conformation.

(A) Superimposed view of all the orientations of all the visible full and partial αK40 loops, showing their orientation. (B) Interaction of K40 loop and DM10 domains from RIB72A (3 domains, blue), RIB72B (3 domains, purple) and CFAP67 (1 domain, cyan). Black box represents the view in (C). (C) Zoom in view of DM10 domains and K40 loop interaction. Asterisk (*) denotes the conserved aromatic residue potentially interact with the K40 loop. (D) Alternative view of DM10 domains interacting with K40 loop. (E) Multiple sequence alignment of DM10 domains from RIB72A, RIB72B and CFAP67. (F) Cryo-EM map (left) and model (right) of the inner junction region of *Tetrahymena* to show the interaction of the full αK40 loop with CFAP52. (G) Cryo-EM map (left) and model (right) of the inner junction region of *Chlamydomonas* to show the interaction of the full αK40 loop with CFAP52. (H) Superimposed view of the *Chlamydomonas* αK40 loop (gray) onto the *Tetrahymena* αK40 loop of B10.

α K40 loop are conserved in its topology (Fig. 2B, C). Sequence alignment of the DM10 domains suggests that an aromatic residue (phenylalanine or tyrosine) is conserved and might be responsible to the interaction with the DM10 domain (Fig. 2C, D, E).

To determine whether the conformation of the α K40 loops is conserved between species, we compared the α K40 loops in the DMTs of *Tetrahymena thermophila* and *Chlamydomonas reinhardtii* (Fig. 2F, G, H). The α K40 loop conformations from B9 and B10 are similar between the two species, and both interact with the conserved CFAP52. These results suggest that the α K40 loop conformation is tuned by its interactions with adjacent MIPs. The interaction between MIPs and the α K40 loop implies that acetylation might play a role in DMT assembly or stability or both.

α K40 acetylation stabilizes the inter-PF angles in the B-tubule

With our observation that acetylated α K40 in the DMT of WT *Tetrahymena* cells adopts a fixed conformation when interacting with MIPs, we evaluated whether the absence of α K40 acetylation affects the MIPs and hence the overall DMT structure. We obtained the DMT structures from the *Tetrahymena MEC17-KO* and *K40R* mutants at 4.5 and 3.5 Å resolution, respectively (Table 1 and 2). In *MEC17-KO*, *MEC17* - an ortholog of the mammalian α -TAT1 α -tubulin K40 acetyltransferase, is knocked out to abolish detectable α K40 acetylation, while in the *K40R* mutant, lysine 40 on ATU1 (the single canonical α -tubulin isotype in *Tetrahymena*) is mutated to arginine to prevent acetylation at the α K40 position (Akella et al., 2010).

Our first observation is that all the decoration of the MIPs appears intact in *MEC17-KO* and *K40R* DMT compared to that of WT (Fig. 3A, Supplementary Fig. 3A, B). This observation is consistent with the phenotypes that the *MEC17-KO* and *K40R* cilia look similar to the WT cilia (Akella et al., 2010; Gaertig et al., 1995). Therefore, our cryo-EM analyses indicate that acetylation does not affect DMT and MIP assembly.

Next, we examined the tubulin structure by comparing the α K60 and α K40 loop structures at PFs A3 and B1 (Fig. 3B-I). Previously, it was reported that α K40 acetylation led to a reduction in the distance of α K60 from the M-loop of the adjacent tubulin from 12 Å to 8 Å (Eshun-Wilson et al., 2019). Our comparison of (acetylated and non-acetylated) α K40 and α K60 loops did not reveal significant structural differences. However, this result does not rule out a change in α K40 and α K60 conformation, which is likely too small to be observed accurately at this resolution.

Since α K40 loops are suggested to be involved in lateral interactions (Cueva, J., et al., 2012; Eshun-Wilson et al., 2019) and acetylated α K40 results in weakened inter-PF interactions (Portran et al., 2017), acetylation might affect inter-PF angles. We measured the inter-PF angles in the DMTs of the WT and the two acetylation-deficient mutants. We observed only minor changes in the inter-PF angles in the A-tubule (Fig. 4A, Supplementary Fig. 4A). However, in the B-tubule, the changes in the inter-PF angles were more prominent (Fig. 4A-M, Supplementary Fig. 4B). Most notably, the change in the inter-PF angle between PFs B7 and B8 and B9 and B10 ranges from 3 to 6 degrees (Fig. 4H-M, Supplementary Fig. 5, Supplementary Table 2). These are significant changes in specific PF curvatures. To make sure that our observation is statistically significant, we analyzed the inter-PF angles using independent reconstructions obtained from two biological replicates of the WT and *K40R* DMT. The measurements of inter-PF angles between the two WT replicates are consistent, suggesting that the inter-PF angles in WT are well-controlled. For two *K40R* replicates, the inter-PF angles have minor changes in the A-tubule (Supplementary Fig. 4A) but for the B-tubule, there are several PF pairs with significant differences (Supplementary Fig. 4B), especially in PFs B7B8 and B9B10. This analysis suggests that the inter-PF rotation angle varies significantly more in acetylation mutants *K40R* and *MEC17-KO* (Supplementary Fig. 4B).

We also inspected the tubulin lattice by measuring the distances between adjacent tubulin dimers. There were small changes in the inter-dimer distances in PFs A1, A3, A5, A8, B3 and B7 between WT and mutants (Supplementary Fig. 4C, Supplementary Table 3). This scale of changes in the

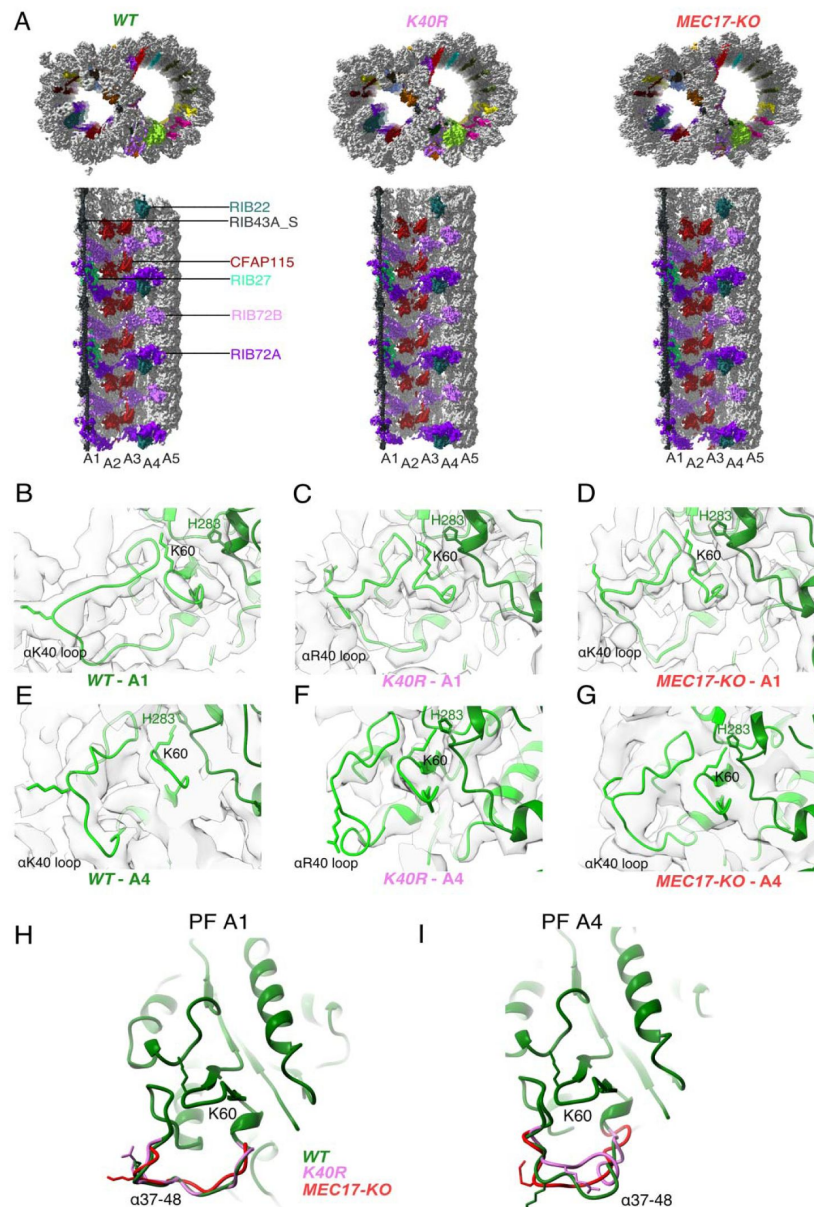


Figure 3

Comparison of DMT structures from WT, MEC17-KO and K40R mutants

(A) Comparison of the cryo-EM density maps of the DMT from WT, K40R, and MEC17-KO strains of *Tetrahymena* to show that the MIPs are intact in all three species. (B-G). Models of the full αK40 loops in PF A1 (B-D) and PF A4 (E-G) from WT, K40R, and MEC17-KO strains. (H-I) Models of the αK40 loops from A1 (H) and A4 (I) superimposed from WT, K40R, and MEC17-KO species.

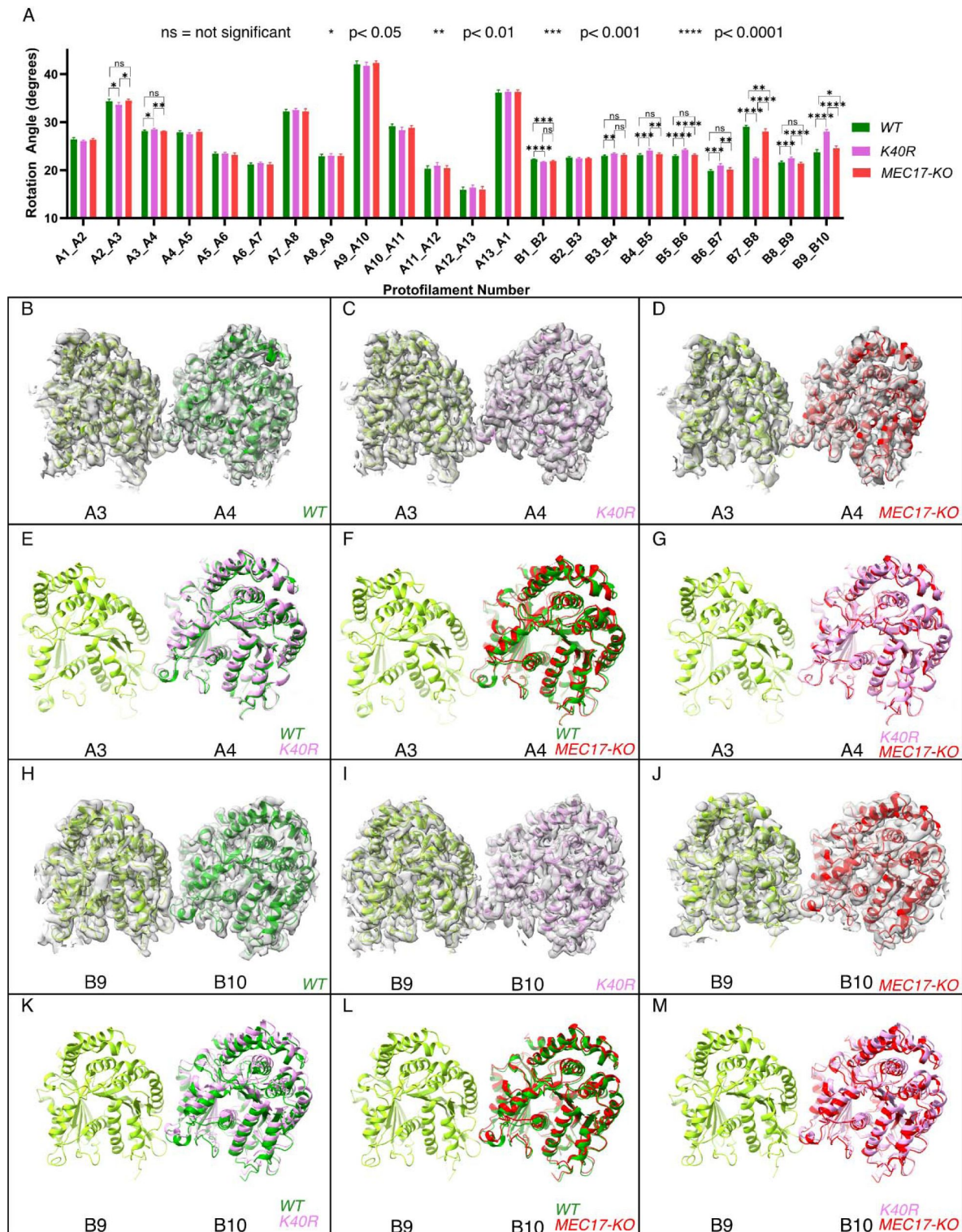


Figure 4

Deacetylation affects the inter-PF angles in the DMT.

(A) Inter-PF rotation angles for each PF across all three strains (WT, K40R, MEC17-KO). (B-G) Comparison of inter-PF rotation angle change between A3 and A4, showing minor changes. (H-M) Comparison of inter-PF rotation angle change between B9 and B10, showing significant changes.

tubulin lattice is significantly smaller (0.3 – 0.5 Å) than the known impacts of the nucleotide state (Zhang et al., 2015 [↗](#)) and missing MIPs (Ichikawa et al., 2019 [↗](#)) (~2 Å difference). This is similar to the observation of differences in interdimer distances in 96% acetylated and 99% de-acetylated microtubules using *in vitro* reconstitution.

Overall, our analysis shows that tubulin lattice changes, specifically inter-PF angles, can be used as an indication of microtubule stability when acetylation is lacking.

Acetylated α K40 loops are less flexible

While we observed some structural differences between acetylated and non-acetylated DMT, our resolution does not allow us to see small changes in the α K40 loops. Therefore, we attempted to detect structural differences using molecular dynamic simulations.

We performed an all-atom molecular dynamic simulation for α -tubulin with acetylated and non-acetylated α K40 to determine whether acetylation changes the loop behavior (Fig 5A, B). To determine how acetylation of α K40 affects the structure of the α K40-loop, we used the K-means clustering method for the α K40-loop region based on the C α position. To avoid bias in clustering, the structures obtained during the simulated trajectories with acetylated and non-acetylated α K40 were mixed, and 10 clusters were created (Fig. 5A, B [↗](#)). Interestingly, clusters comprise mostly acetylated or non-acetylated conformations but not mixed populations. This indicates that acetylated and non-acetylated α K40 loops adopt distinctively different conformations. In addition, the most populated cluster consists of ~50% of all acetylation conformations, while the non-acetylated α K40 loops form five clusters of less than 30% conformations. Therefore, the acetylated α K40 loop adopts more rigid conformations than non-acetylated α K40 loop. It appears that this finding is similar to that of a previous study (Eshun-Wilson et al., 2019 [↗](#)) despite differences in molecular dynamic setups of porcine and ciliate tubulins and clustering methods.

Lack acetylation of α K40 does not significantly affect tubulin and MIP interactions

Since we did not observe any obvious structural changes in the decoration of the MIPs in the cryo-EM maps of the 48-nm repeat of *MEC17-KO* and *K40R* mutants compared to WT (Fig. 3A [↗](#), Supplementary Fig. 3), non-acetylated α K40 does not appear to significantly weaken the binding of MIPs. Therefore, we further explored the degree to which the interaction between the α K40 loop and MIP is affected by acetylation using coarse-grain molecular dynamic simulation to compare the energy between tubulins and CFAP52 with the acetylated α K40 and the non-acetylated α K40 structures (Fig. 5C [↗](#)).

We found that both acetylated and non-acetylated α K40 have a stabilizing effect when binding to CFAP52. On the other hand, when calculating the mean and standard error of the energy between tubulins and CFAP52 from the entire trajectory, the energy of the acetylated α K40 case is only slightly lower than that of the non-acetylated α K40 case (Fig. 5D [↗](#)). Similarly, no significant difference between acetylated and non-acetylated α K40 in interactions with similar setup using RIB72A. Even when accounted for the periodic structures of DMT with many MIPs, the energy difference between acetylated and non-acetylated α K40 with MIPs is still not significant.

Our coarse-grain molecular dynamic simulations of MIPs and α K40 suggest that acetylation likely does not play a significant role in the interaction with MIPs.

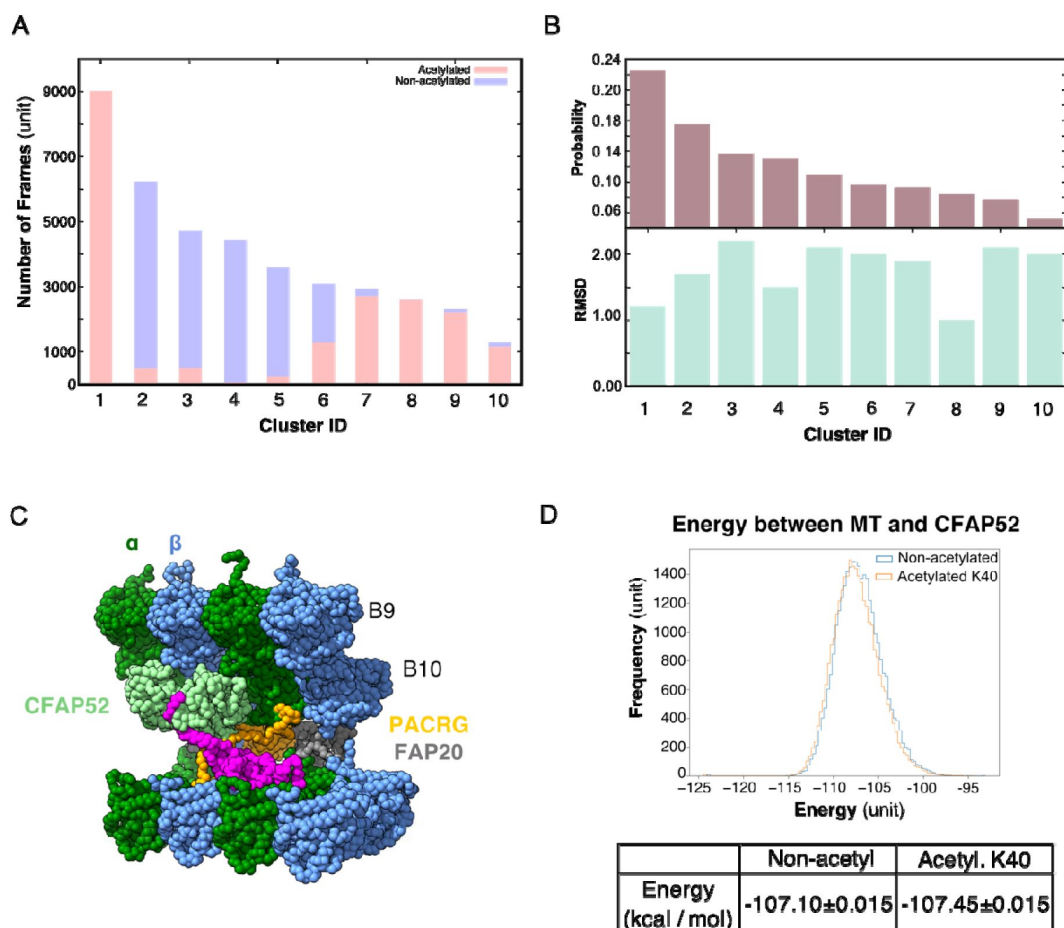


Figure 5.

Molecular dynamic simulations of the acetylated and non-acetylated α K40 loops.

(A) All-atom simulations of α K40 loop clusters in different conformations of acetylated (pink) and base/non-acetylated (blue) indicate that acetylated conformations adopt higher frames and are less flexible. (B) RMSD and probability of each cluster simulated in A. (C) Molecular dynamics coarse grain model of the inner junction region of *Tetrahymena*; each amino acid is 1 bead. (D) Graph showing the energy difference (in kcal/mol) between base (non-acetylated) and acetylated α K40 to show that each acetylated α K40 has slightly lower energy than the non-acetylated α K40.

Mass spectrometry reveals changes in DMT protein composition in response to a lack of α K40 acetylation

Our analysis thus far has failed to reveal significant structural differences in the MIPs within the DMT structures of *WT*, *K40R* and *MEC17-KO* mutants. Therefore, we searched for more subtle changes in the axoneme composition using mass spectrometry. We analyzed the same DMT samples used for cryo-EM, which contain no membrane and matrix fractions, to see any proteomic changes due to the lack of acetylation. In addition, to eliminate the downstream effect of the lack of acetylation i.e. destabilization of DMT, we combined the mass spectrometry results in this study with the mass spectrometry studies of the *RIB72A/B-KO* and *RIB72B-KO* mutants (Kubo et al., 2023 [↗](#)). The *RIB72A/B-KO* mutant lacks both RIB72A and RIB72B, which leads to a significant number of MIPs missing and slower swimming speed (Stoddard et al., 2018 [↗](#)). Therefore, we can use *RIB72B-KO* and *RIB72A/B-KO* as controls to specifically look for the upstream effect due to the lack of acetylation.

We first performed a control of proteins not supposed to interact with α K40 by analyzing the levels of radial spoke proteins and found no significant differences (Supplementary Fig. 6A). We then looked at the MIP abundance. Most MIPs showed no significant changes in their abundance (less than 1.5-fold changes) except for CFAP112, CFAP141 and RIB27 (Fig. 6A [↗](#)) in the *K40R* and *MEC17-KO* samples. However, we did not observe any differences in the periodicity of CFAP112, CFAP141 and RIB27 in the cryo-EM maps of the DMTs from *WT*, *K40R* and *MEC17-KO* mutants. As shown recently in the case of CFAP77A and CFAP77B, certain MIPs might not localize consistently along the length of the cilium (Kubo et al., 2023 [↗](#)), and there might be some changes in the occupancy of those MIPs in a specific region that lead to changes in their abundance.

There were 11 proteins significantly elevated in both *K40R* and *MEC17-KO* mutants by fourfold compared to the *WT* cilia (Fig. 6B [↗](#), Supplementary Fig. 6B, C, Supplementary Table 4 and 5). One of these proteins was also elevated in the *RIB72A/B-KO* mutant vs. *WT*, and therefore, 10 proteins were specifically upregulated in both the *K40R* and *MEC17-KO* strains (Fig. 6B, G [↗](#)). Similarly, we found that three proteins were significantly reduced in both *K40R* and *MEC17-KO* but not in the *RIB72A/B-KO* knockout mutant (Fig. 6C, E [↗](#) and Supplementary Table 6 and 7). Among the proteins reduced or missing in the *K40R* and *MEC17-KO* mutants, there are protein phosphatase 2A-related proteins: PP2A regulatory subunit A (TTHERM_00766530, UniProt ID: I7MAR7), PP2C (TTHERM_00316330, UniProt ID: I7LW71), and PP2A (TTHERM_00355160, UniProt ID: Q22Y55) (Fig. 6D [↗](#)). Furthermore, one kinase was downregulated in *MEC17-KO* and *K40R* cells (Fig. 6E [↗](#)) (TTHERM_00623090, UniProt ID: Q240X5). In *Chlamydomonas*, PP2A is present in DMT and required for normal ciliary motility (Elam et al., 2011 [↗](#)). It was previously reported that the knockdown of PPP1R2 (protein phosphatase inhibitor 2) reduces α K40-acetylation in the primary cilium of human retinal epithelial cells (Wang & Brautigan, 2008 [↗](#)). Inhibition of protein phosphatase (1 and 2A) with calyculin in PPP1R2 knockdown cells partially rescued the acetylation of ciliary microtubules. These results suggest that the lack of α K40 acetylation reduces PP2A activity in DMT. Therefore, we can infer that acetylation and phosphorylation interact in DMT.

Discussion

In this work, we reinforced the notion that acetylation of α K40 makes the loop less flexible (Eshun-Wilson et al., 2019 [↗](#)), which is observed by molecular dynamics. In addition, we performed structural characterization of the DMTs from two acetylation mutants, *K40R* and *MEC17-KO*, and compared them to that of the *WT*. We showed that the α K40 loops are structured when they interact with MIPs and identified DM10 as an α K40 interacting domain. This suggests that the α K40 loop plays an important role in certain MIP-tubulin interactions. On the other hand, the α K40 loop-

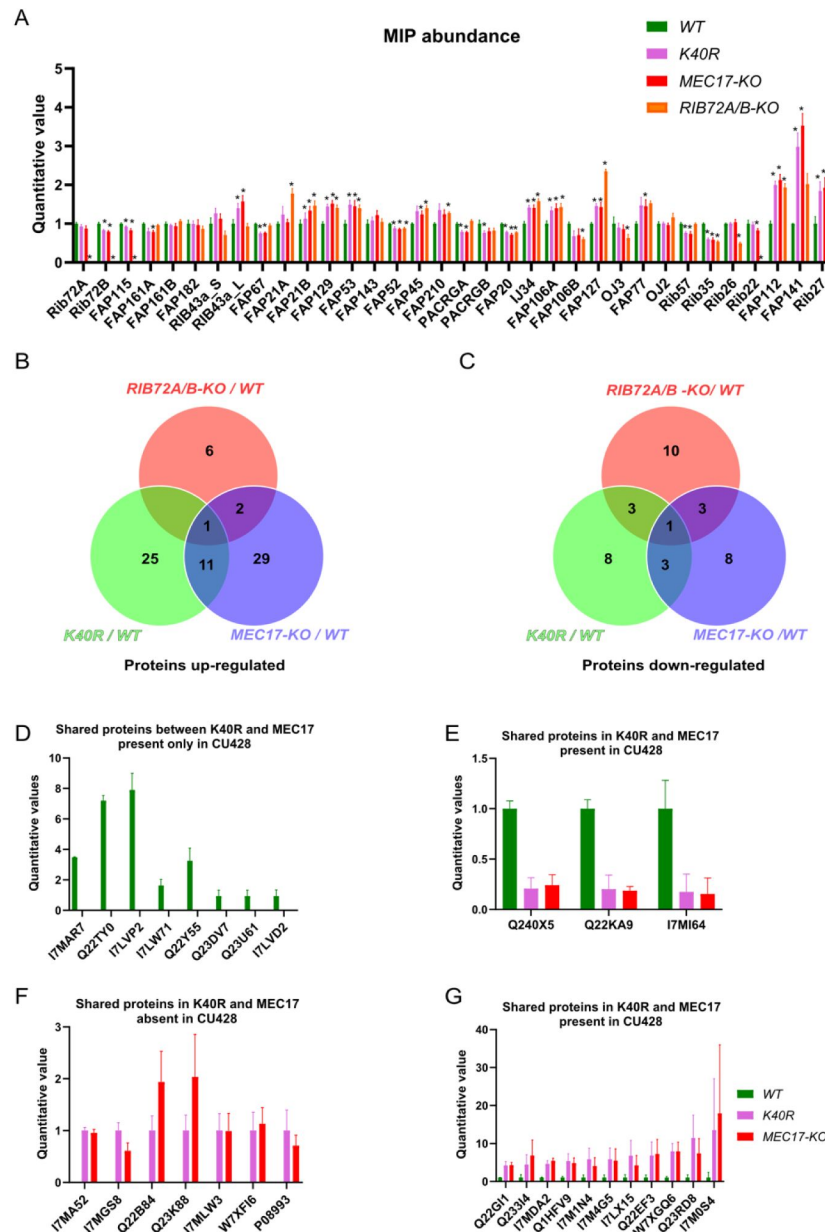


Figure 6.

Mass spectrometry of WT, K40R and MEC17-KO mutants.

(A) Bar graph showing the abundance of MIPs based upon quantitative values (normalized total spectra) from mass spectrometry. Asterisk (*) indicates a significant difference with $p < 0.05$. (B) Proteins upregulated in RIB72A/B, K40R, and MEC17-KO mutants compared with the WT. (C) Proteins downregulated in RIB72A/B, K40R, and MEC17-KO mutants compared to the WT. (D) Proteins only found in the mass spectrometry of WT when compared with K40R and MEC17-KO mutants. (E) Downregulated proteins in both K40R and MEC17-KO mutants compared to WT. (F) F. Proteins in both K40R and MEC17-KO mutants but are absent in WT. (G) Upregulated proteins in both K40R and MEC17-KO mutants compared to WT.

MIP interactions do not detectably change their structures between acetylated and non-acetylated α K40. These results imply that a complete DMT with MIPs is assembled without the need of acetylation. Later, the MEC17/ATAT1 acetyltransferase acetylates α K40 within the DMT.

Interestingly, we observed significant changes in inter-PF angles in the B-tubule, where MIPs are fewer. Our results suggest that α K40 loop interactions with MIPs are the dominant interactions that stabilize microtubules regardless of acetylation status. When there are fewer MIPs and thus less interaction between the α K40 loop and MIPs, the contribution of the acetylation of α K40 to the lateral interaction between adjacent PFs becomes more significant (**Fig. 7**). As a result, the lack of acetylation destabilizes DMT and leads to tubulin lattice alteration and instability. Our results agree with an *in vitro* study showing that α K40 acetylation affects inter-PF interaction and, as a result, improves mechanical properties of reconstituted microtubules (Portran et al., 2017). In a recent study (Viar et al., 2023), the tip region of the *Chlamydomonas* cilia is acetylated early during growth compared to other PTM such as polyglutamylation and polyglycylation. In the tip, the microtubules exist as singlet microtubules and do not have regular MIP binding pattern like in the base (Legal et al., 2023). As a result, acetylation might play an essential role in the ciliary tip region to stabilize microtubules. Moreover, our study supports the notion that acetylation of tubulin is not a biphasic switch but a fine-tuning mechanism that impacts microtubule stability. Our results again demonstrate that the tubulin lattice can be a read-out for DMT stability (Ichikawa et al., 2019). Recently, it has been shown that acetylation of K394, which is located at the α - β -tubulin dimer interface, is specific to flies' nervous system and is critical to neuronal growth (Saunders et al., 2022). This finding might suggest that different acetylation sites can be fine-tuned for different purposes.

Since both acetylation mutants lack protein phosphatase 2A and its regulatory subunits, there could be an interaction between tubulin acetylation and protein phosphatase 2A in cilia. TGF- β -activated kinase 1 (TAK1) is an important activator of α TAT1 in mice (Shah et al., 2018). We can speculate that the absence of α K40 acetylation could also affect the enzyme MEC17/ATAT1 and its availability for phosphorylation by TAK1 or other kinases. Possibly, the lack of α K40 acetylation changes the phosphorylation levels by altering the balance between kinases and phosphatases (protein phosphatase 2A in this case). Interestingly, HDAC6 deacetylase is also regulated by multiple kinases (Du et al., 2015). This again highlights the interdependence of phosphorylation and acetylation in cilia. Furthermore, crosstalk occurs between α K40 acetylation and other tubulin PTMs, namely polymodification. For example, *Tetrahymena* mutants lacking tubulin polyglycylation (TTLL3) have elevated levels of α K40 acetylation (Dorota Wloga et al., 2009). Our mass spectrometry results showed that a tetratricopeptide repeat protein (TTHERM_00313720, UniProt ID: Q22KA9) was downregulated in MEC17-KO and K40R mutants (**Fig. 6E**). Protein BLAST found a homolog in *Homo sapiens* as tetratricopeptide repeat protein 30B/IFT70 (UniProt ID: Q8N4P2), an IFT protein whose depletion reduces polyglutamylation of axonemal tubulin (Pathak et al., 2007). We also observed that the absence of α K40 acetylation affects the PF angles, and the impact is stronger on the B-tubule. The B-tubule is enriched in polymodification (polyglutamylation and polyglycylation), and therefore, the altered B-tubule structure could affect the access of enzymes that add or remove polymodification. In fact, some of the effects of the absence of α K40 acetylation based on the use of non-acetylated axonemes (Reed et al. 2006) could be due to alterations of the B-tubule, namely, the levels of polymodification. Together with our results and others (Akella et al., 2010; Kubo et al., 2015; Rogowski et al., 2010b; Wloga et al., 2008; Dorota Wloga et al., 2009), these results suggest that the PTM in cilia is in a balancing act. Changing one type of PTM can shift the balance of other types of PTMs.

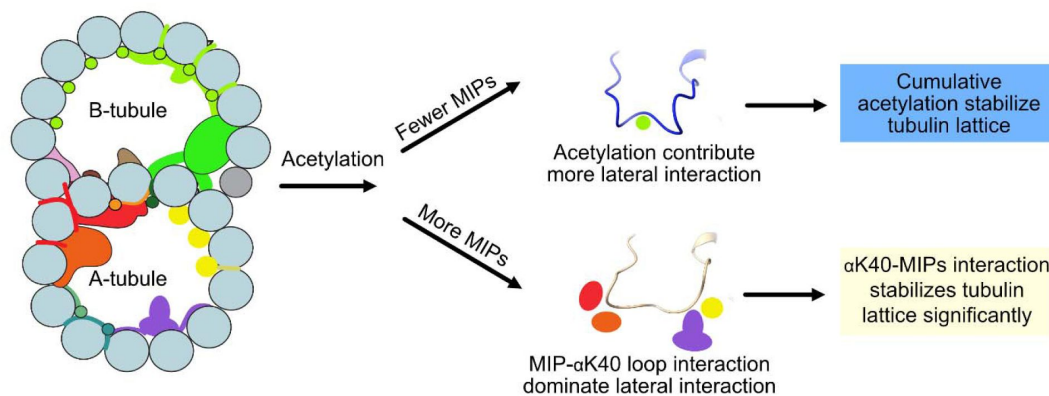


Figure 7

Models of acetylation contribution in the DMT

In the case of more MIPs, such as in the A-tubule, the MIP- α K40 interaction dominates the contribution to the lateral interaction; therefore, deacetylation does not affect the structures significantly. With fewer MIPs, such as in the B-tubule, acetylation contribution to the lateral interaction becomes significant and therefore can contribute to the stabilization of the tubulin lattice.

Materials and methods

Growth of *Tetrahymena* cells for isolation

Bean media (Williams et al., 1980 [DOI](#)) was used to store *Tetrahymena* cells (WT (CU428), K40R, MEC17-KO), with 4 μ L of the cell culture transferred into 40 mL of liquid SPP media (Gorovsky et al., 1975 [DOI](#)). The cells were grown for 7 days at room temperature and then transferred into 50 mL of SPP liquid media overnight growth at 30°C with 150 RPM shaking in a Thermo Fisher MAXQ8000 incubator. Then, 50 mL of this overnight culture was added to 900 mL of liquid SPP media and grown at 30°C (MAXQ8000) at 150 RPM for two days or until the OD₆₀₀ was 0.7.

Flagella isolation via dibucaine treatment

To harvest *Tetrahymena* cells, overnight culture was centrifuged at 700 $\times$ g for 10 minutes with slow deceleration at 22°C in an Avanti Centrifuge (Rotor JLA-8.1000). Ten milliliters of room temperature SPP medium supplemented with DTT was used to resuspend the cell pellet and then adjusted to a final volume of 24 mL, followed by transfer to an ice-cold 250 mL Erlenmeyer flask. Immediately, 1 mL of dibucaine (dissolved in distilled water at 25 mg/mL) was added to the flask and gently swirled for exactly 60 seconds in an ice water bath. To stop the reaction, 75 mL of ice-cold SPP media (supplemented with 1 mM EGTA) was immediately added to the Erlenmeyer flask and then split into two 50 mL conical tubes for centrifugation at 2000 $\times$ g for 10 minutes at 4°C with no deceleration (Sorvall ST 16R, Rotor 75003181). The supernatant that contained cilia was transferred to centrifuge tubes for the Beckman Coulter JA 25.50 rotor, approximately 30 mL per tube, for centrifugation at 17000 $\times$ g for 40 minutes at 4°C with slow deceleration (Avanti, Rotor JA25.50). The pellet was gently washed with Cilia Wash Buffer (50 mM HEPES at pH 7.4, 3 mM MgSO₄, 0.1 mM EGTA, 1 mM DTT, 250 mM sucrose), and frozen with liquid nitrogen for storage in a -80°C freezer.

Purification of DMT fraction

The cilia suspension was thawed on ice and then centrifuged at 16000 g and 4°C for 10 minutes in a microfuge in a refrigerated room (Eppendorf, Centrifuge 5415 D). The pellet was resuspended in 250 μ L of ice-cold Cilia Final Buffer (50 mM HEPES at pH 7.4, 3 mM MgSO₄, 0.1 mM EGTA, 1 mM DTT, 0.5% trehalose, 1 mM PMSF). To clean the cilia, the resuspended cilia were centrifuged at 16000 $\times$ g for 10 minutes at 4°C in a microfuge (Eppendorf, Centrifuge 5415 D), and the supernatant was removed. Then, the pellet was resuspended in 250 μ L of Cilia Final Buffer [without trehalose but with 44.1 μ L of 10% NP-40 alternative (Millipore Sigma, 492016)] added to a final concentration of 1.5% NP-40]. The sample was placed on ice to incubate for 30 minutes before the demembrated flagella supernatant was removed after centrifugation at 16,000 $\times$ g for 10 min at 4°C in a microfuge (Eppendorf, Centrifuge 5415 D). The pellet containing the axoneme was resuspended in 245 μ L of Cilia Final Buffer (without trehalose), and then 2.5 μ L of 100 mM ADP (to a final concentration of 1 mM ADP) was added for incubation at room temperature for 10 minutes. This was followed by adding 2.5 μ L of 10 mM ATP (to a final concentration of 1 mM ATP) for incubation at room temperature for 10 minutes. Bradford reagent (Bio-Rad 5000201) was used to measure the total protein concentration, and the final protein concentration was adjusted to 3 mg/mL using Cilia Final Buffer (without trehalose).

Cryo-EM sample preparation

The concentration of the DMT solution was adjusted to 3 mg/mL. Quantifoil R2/2 grids (Electron Microscopy Sciences, #Q225CR-06) were treated using 1 mL of chloroform overnight followed by negative glow discharge (30 seconds at 25 mA). Then, 3.5 μ L of axoneme sample was applied to treated grids inside the Vitrobot Mark IV (Thermo Fisher) at a blot force of 3, blot time of 5 seconds, and drain time of 0.5 seconds, followed by plunge freezing into liquid ethane.

Cryo-EM data acquisition

Using a Titan Krios 300 kV FEG electron microscope (Thermo Fisher) with a K3 Summit direct electron detector (Gatan, Inc.) and the BioQuantum energy filter (Gatan, Inc.), movies of the axoneme were acquired at 64kx nominal magnification (calculated pixel size of 1.370 Å/pixel) using SerialEM (Mastronarde, 2005). A total dose of 45 electrons per Å² over 40 frames for the *WT* and *MEC17-KO* datasets. A total dose of 73 electrons per Å² per frame over 30 frames for the *K40R* dataset. The defocus range was between -1.0 and -3.0 μm at an interval of 0.25 μm.

Cryo-EM image processing

Motion correction and dose-weighting of the movies were performed using MotionCor2 (Zheng et al., 2017 [↗](#)) implemented in Relion 3 (Zivanov et al., 2018 [↗](#)), and the contrast transfer function parameters were estimated using Gctf (Zhang, 2016 [↗](#)). Micrographs with apparent drift, ice contamination, and poor contrast transfer function estimation were discarded (18384, 25610, 4283 micrographs for *WT*, *K40R* and *MEC17-KO* data, respectively). The filaments were picked manually using e2helixboxe (Tang et al., 2007 [↗](#)).

An 8-nm periodicity was used to pick particles of 512 x 512 pixels, binned twice, and prealigned using a modified version of the Iterative Helical Real Space Reconstruction script (Egelman, 2007 [↗](#)) in SPIDER (Frank et al., 1996 [↗](#)) to work with nonhelical symmetry. The alignment parameters were then transferred to FREALIGN to align the particles for 6 iterations in FREALIGN (Grigorieff, 2007 [↗](#)) and then converted into Relion 3.0. In Relion 3, iterative per-particle-defocus refinement and Bayesian polishing were performed for the 80 nm particles.

Particles were subtracted from their tubulin lattice signal and underwent 3D classification into two classes to obtain the 16-nm repeat particles. The 16-nm repeat particles were then subjected to 3D classification into 3 classes to obtain the 48-nm repeat particles. The 48-nm particles were then refined, resulting in resolutions of 3.5 and 4.3 Å for *K40R* and *MEC17-KO* DMTs from 182387 and 39417 particles, respectively.

To improve the local resolution for each PF during modeling, we performed focused refinements by using masks to cover adjacent PF regions in the DMT. Next, the maps were enhanced by DeepEnhancer (Sanchez-Garcia et al., 2021 [↗](#)) to improve visualization and interpretability.

Tubulin Modeling

The *WT* tubulin model 6U0H (Ichikawa et al., 2019 [↗](#)) was first fitted into the higher-resolution *K40R* cryo-EM map and then locally modeled using Coot (Emsley et al., 2010 [↗](#)) and real space refined in Phenix (Adams et al., 2010 [↗](#)) for generation of the *K40R* tubulin model. The *WT* and *MEC17-KO* tubulin models were generated by fitting the *K40R* tubulin model in *WT* and *MEC17-KO* cryo-EM maps, respectively, followed by refinement in Coot (Emsley et al., 2010 [↗](#)) and Phenix (Adams et al., 2010 [↗](#)). The αK40 loop regions from *WT*, *K40R*, and *MEC17-KO* were locally modeled in focused refinement cryo-EM maps using Coot (Emsley et al., 2010 [↗](#)) and Phenix (Adams et al., 2010 [↗](#)). All the maps and model visualization were taken using ChimeraX (Goddard et al., 2018 [↗](#)).

Coarse-grained molecular dynamic simulation

Based on the atomic structures of the four tubulin dimers, CFAP52 and IJ34, we performed coarse-grained molecular dynamic simulations. The purpose of this simulation was to check the effect of acetylated αK40 on the binding stability of CFAP52. In the coarse-grained model, each amino acid was represented as a single bead located at its Cα position. To observe their dynamics, we used the excluded volume effect, electrostatic interaction, and the energy function AICG2+ (Li et al., 2012 [↗](#); Li et al., 2014 [↗](#)). In AICG2+, the reference structure was assumed to be the most stable conformation, and their parameters were modified from the reference. It is known that the

intradimer interaction is much stronger than the interdimer interaction, and the interdimer interaction is much stronger than the intra-PF interaction, so we set the interdimer and PFs' nonlocal native interaction force to 0.8 and 0.3 times the original value, respectively, while that of the intradimer was kept as the original value (1.0 times). Of note, three residues (PHE133, GLY308, and GLU401) of the B9-PF beta-tubulin at the plus end side were anchored in their position for convenience analysis. We performed the simulation 10 times with acetylated α K40 and non-acetylated α K40 structures using the CafeMol package version 2.1 (Kenzaki et al., 2011 [DOI](#)). Each molecular dynamic simulation took 3×10^7 molecular dynamic steps, and they were conducted by the underdamped Langevin dynamics at 300 K temperature. We set the friction coefficient to 0.02 (CafeMol unit), and default values were used for others.

Normally, in dealing with electrostatic interactions, LYS and ARG, GLU and ASP, and other amino acids were given a charge of +1, -1, and 0, respectively. However, in this simulation, it was necessary to evaluate the electrostatic interaction as accurately as possible, so we calculated the surface charge density from the all-atom structure and remapped the charge distribution using only Ca beads to reproduce the all-atom surface charge distribution. The technique is called RESPAC (Terakawa & Takada, 2014 [DOI](#)). We applied RESPAC to regions without missing data, such as the α K40-loop and E-hook. For the missing region (and so we performed loop modeling by MODELLER (Šali & Blundell, 1993 [DOI](#)), we treat their charge by default definition. In the coarse-grained model, each amino acid is treated as a single bead, so we simply assumed that if α K40 was acetylated, its charge was zero, while non-acetylated α K40 had a +1 charge.

All-atom molecular dynamic simulation

The purpose of the all-atom molecular dynamic simulation was to check whether the acetylated α K40 loop takes fewer conformations than the non-acetylated α K40 loop. In the all-atom molecular dynamic simulation using GROMACS (Abraham et al., 2015 [DOI](#); Pronk et al., 2013 [DOI](#)), we used the GROMOS54a7 force field for protein (Schmid et al., 2011 [DOI](#)) and SPC for solvent water (Jorgensen & Tirado-Rives, 2005 [DOI](#)). We added sodium and chloride ions to neutralize the system and to make the salt concentration approximately equal to 0.1 M. Energy minimization by the steepest descent minimization algorithm was followed by equilibration with NVT and NPT for 100 ps at 300 K. In the production run, we used NPT ensemble simulations with 1 atm and 300 K. The production run consisted of a 1 fs step for 180 ns. The α -tubulin at the plus end of B9-PF was used as a reference structure for the simulation. To model the structure of acetylated α K40, Vienna-PTM 2.0 was used (Margreitter et al., 2013 [DOI](#); Margreitter et al., 2017 [DOI](#); Petrov et al., 2013 [DOI](#)). The force field used (GROMOS54a7) had already been set up for the acetylated lysine.

Measurement of inter-PF rotation angles

The inter-PF rotation angle can be defined by the lateral rotation angle between each subsequent PF pair. The rotation angles and Z-shift between PF pairs were measured using the “measure” command from ChimeraX (Pettersen et al., 2004 [DOI](#)). Data for WT, K40R, and MEC17-KO cells were compiled into GraphPad Prism 9 to perform ANOVA and plotting.

To perform statistical analysis of the rotation angles, we performed independent cryo-EM reconstructions of two WT datasets (WT G1 and G2) and two K40R datasets (K40R G1 and G2) (Supplementary Fig. 3A, B). These datasets represent biological duplicates with cilia from cells cultured on different dates and prepared for cryo-EM on different dates. Unpaired t tests or Mann Whitney tests were carried out for each angle between both WT and K40R groups separately. One-way ANOVA or Kruskal Wallis tests were carried out for each angle between combined WT points, combined K40R points and MEC17-KO.

Measurement of interdimer distance

We docked in the atomic models of the α - and β -tubulins in the maps of *WT*, *K40R*, and *MEC17-KO*. The interdimer distance was measured between the N9 GTP of α -tubulin and that of the next α -tubulin in the same PF using the “distance” command from Chimera (Pettersen et al., 2004 [DOI](#)).

Mass Spectrometry

Samples prepared for cryo-EM were used for mass spectrometry analysis. Laemmli buffer at 4X (#1610747, Bio-Rad) was added to the microtubule fraction samples in Cilia Final Buffer buffer so that it was 1x, and 25–30 μ g protein was loaded on the SDS–PAGE gel. Electrophoresis was performed, but the run was terminated before the proteins entered the separation gel. A band containing all proteins in the sample was then cut out from the gel and subjected to in-gel digestion. The obtained peptides ($\sim$ 2 μ g) were chromatographically separated on a Dionex Ultimate 3000 UHPLC. First, peptides were loaded onto a Thermo Acclaim Pepmap (Thermo, 75 μ m ID $\times$ 2 cm with 3 μ m C18 beads) precolumn and then onto an Acclaim Pepmap Easyspray (Thermo, 75 μ m $\times$ 25 cm with 2 μ m C18 beads) analytical column and separated with a flow rate of 200 nL/min with a gradient of 2–35% solvent (acetonitrile containing 0.1% formic acid) over 2 hours. Peptides of charge 2+ or higher were recorded using a Thermo Orbitrap Fusion mass spectrometer operating at 120,000 resolution (FWHM in MS1, 15,000 for MS/MS). The data were searched against the *Tetrahymena thermophila* protein dataset from UniProt (<https://www.uniprot.org/> [DOI](#)).

Mass spectrometry data were analyzed by Scaffold_4.8.4 (Proteome Software Inc.). Proteins with mean values of exclusive unique peptide count of 2 or more in the *WT* mass spectrometry results were used for analysis. Raw mass spectrometry data were normalized by the total spectra. ANOVA statistical tests were applied to *MEC17-KO*, *K40R* and *WT* mass spectrometry results using biological triplicates. Proteins exhibiting a minimum of twofold increase/decrease and a statistical significance threshold ($p < 0.05$) in mutants compared to *WT* were identified as up- or downregulated.

Data availability

The data produced in this study are available in the following databases:

- Cryo-EM maps of the 48-nm repeat of *MEC17-KO* DMT: EMD-40436
- Model coordinates of the *MEC17-KO* DMT: PDB 8SF7

The *WT* and *K40R* structures used in this paper have associated maps in the Electron Microscopy Data Bank database with the following EMD IDs EMD-29685 and EMD-29692.

The *WT* and *K40R* structures used in this paper have coordinates in the RCSB Protein Data Bank database with the following PDB IDs 8G2Z and 8G3D.

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Editors

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University of California, Davis, United States of America

Senior Editor

Qiang Cui

Boston University, Boston, United States of America

Reviewer #1 (Public Review):

Summary:

The study "Effect of alpha-tubulin acetylation on the doublet microtubule structure" by S. Yang et al employs a multi-disciplinary approach, including cryo-electron microscopy (cryo-EM), molecular dynamics, and mass spectrometry, to investigate the impact of α -tubulin acetylation at the lysine 40 residue (α K40) on the structure and stability of doublet microtubules in cilia. The work reveals that α K40 acetylation exerts a small-scale, but significant, effect by influencing the lateral rotational angle of the microtubules, thereby affecting their stability. Additionally, the study provided an explanation of the relationship between α K40 acetylation and phosphorylation within cilia, despite that the details still remain elusive. Overall, these findings contribute to our understanding of how post-translational modifications can influence the structure, composition, stability, and functional properties of important cellular components like cilia.

Strengths:

- (1) Multi-Disciplinary Approach: The study employs a robust combination of cryo-electron microscopy (cryo-EM), molecular dynamics, and mass spectrometry, providing a comprehensive analysis of the subject matter.
- (2) Significant Findings: The paper successfully demonstrates the impact of α K40 acetylation on the lateral rotational angles between protofilaments (inter-PF angles) of doublet microtubules in cilia, thereby affecting their stability. This adds valuable insights into the role of post-translational modifications in cellular components.
- (3) Exploration of Acetylation-Phosphorylation Relationship: The study also delves into the relationship between α K40 acetylation and phosphorylation within cilia, contributing to a broader understanding of post-translational modifications.
- (4) High-quality data: The authors are cryo-EM experts in the field and the data quality presented in the manuscript is excellent.
- (5) Depth of analysis: The authors analyzed the effects of α K40 acetylation in excellent depth which significantly improved our understanding of this system.

Weaknesses:

I have no major concerns about this paper.

<https://doi.org/10.7554/eLife.92219.2.sa0>

Author Response

The following is the authors' response to the original reviews.

We would like to thank the reviewer for the constructive comments. We have revised the papers to address the concerns. In summary, here is what we included in the revised version.

- Statistical analysis using biological replicate datasets for WT and K40R doublet microtubule.
- Addition figures for statistical analysis and MIP decorations in MEC17-KO and K40R.
- Revised texts and figures to reflect the new changes, cite proper references and fix small errors throughout the text.

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(4) High-quality data: The authors are cryo-EM experts in the field and the data quality presented in the manuscript is excellent.

(5) Depth of analysis: The authors analyzed the effects of α K40 acetylation in excellent depth which significantly improved our understanding of this system.

Thank you for highlighting the strength of our paper.

Weaknesses:

I have no major concerns about this paper, but would recommend that a few minor issues be addressed.

(1) Lack of Statistical Details: The review points out that the paper could benefit from providing more statistical details, such as the number of particles and maps used for analysis, randomization methods, and dataset splitting for statistical analyses.

To address this, we analyzed the true biological replicate datasets (different cultures, cryo-EM vitrification and data collection) from WT and K40R. Since the MEC17-KO was collected as only one dataset, we decided to not divide the MEC-17 using randomization since the division does not lead to independent sets, which tends to yield identical results in the case of cryo-EM. The biological replicates help us to see how consistent is our structure data for interpretation. The information about the replicate dataset is now included in Table 1. The description of the analysis is highlighted in the manuscript and included in the Materials & Methods and Fig. S4.

In summary, the biological replicate between the WT data indicates that the inter-PF rotation angles are significantly consistent between two biological replicates. On the other hand, there are variations in the inter-PF angles between two replicates of K40R data in the B-tubule (Fig. S4B).

Overall, when pooling the data together (6 + 6 measurement points for WT dataset 1 & 2 and 6 + 6 measurement points for K40R dataset 1 & 2 and 6 measurement points for MEC17-KO) (Fig. S4), our analysis yields the same statistical significance as the average of all datasets (6 measurement points of the total averages for WT, K40R and MEC17-KO) (Fig. 3).

In addition, the variation in inter-PF rotation angles between certain PF pairs within the K40R replicates (B7B8 and B9B10) is similar to the variation to MEC17-KO. This suggests that the deacetylation induces variation in inter-PF angles while the inter-PF angles are maintained consistently in WT.

(2) Questionable Conclusion Regarding MIPs: The reviewer suggests caution in the paper's conclusion that "Acetylation of α K40 does not affect tubulin and MIPs." The reviewer recommends that this conclusion be more specific or supported by additional evidence to exclude all other possibilities.

We now revised the text to make sure we do not overclaim that "Acetylation of α K40 does not affect tubulin and MIPs." We now describe more specifically as "Lack acetylation of α K40 does not significantly affect tubulin and MIP interactions". Also the text was edited to make the statement more specific.

(3) Need for Additional Visual Data: The reviewer recommends that an enlarged local density map along with fitted PDB models be provided in a supplementary figure, such as Figure 4.

We now include the density maps and fitted PDB models in Fig. 4 and Fig. S5. We also include more snapshots of the MIP in K40R and MEC17-KO in Figure S3.

Overall, the paper is strong in its scientific approach and findings but could benefit from additional statistical rigor and clarification of certain conclusions.

Page 11, Line 226: "cluster consists of only ~ acetylated", lacks the percentage. Please correct this.

We corrected it.